Sally Mohammadzadeh

Computer Engineering Student | AI & Data Engineering

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PROFILE

Computer Engineering student at Politecnico di Torino targeting AI and data engineering internships. Computer Vision Engineer on DIANA's 4th-place team among 126 teams at ERC 2025. Built end-to-end machine learning and data systems with Python, YOLOv8, Docker, PostgreSQL, and Streamlit.

EDUCATION

Politecnico di Torino

Bachelor's Degree in Computer Engineering | Sep 2022 - Present | Turin, Italy

Relevant coursework: Databases, Cybersecurity, Algorithms & Data Structures, OOP, Computer Architecture

EXPERIENCE

Computer Vision Engineer - DIANA Student Robotics Team

Politecnico di Torino | Oct 2023 - Present

- Contributed computer vision work to DIANA's 4th-place finish among 126 teams at the European Rover Challenge (ERC) 2025.
- Built a YOLOv8 object-detection pipeline for probe detection, covering field data collection, CVAT annotation and quality control, augmentation, model training in Python, and OpenCV-based development.
- Coordinated the computer vision workstream with mechanical and software subteams under competition deadlines.
- Learned MATLAB and Simulink independently to support inverse-kinematics simulation for the rover arm.

PROJECTS

SpaceX Launch Data Pipeline

Personal Project | 2026 | github.com/sallyze/spacex-pipeline

- Built a Dockerized 4-stage ETL pipeline that ingests 205 SpaceX launches from a REST API, transforms and aggregates data with Pandas, and persists results in two PostgreSQL 15 relational tables.
- Built a Streamlit dashboard that queries PostgreSQL directly for KPI metrics and yearly launch-success charts; the full stack starts with one docker compose up command.

Triple Data Encryption Standard (3DES) Encryption

University Project | 2025 | github.com/sallyze/cryptography-Triple-DES

- Implemented 3DES encryption and decryption from scratch in pure Python without external libraries, including S-boxes, Feistel structure, initial/final permutations, and 48-round key scheduling across three keys.
- Added ECB mode, optional PKCS-style padding, and a written technical report.

SKILLS

Programming: Python, C, SQL, MATLAB
ML & Computer Vision: PyTorch, NumPy, Pandas, YOLOv8, OpenCV, CVAT
Data & Tools: PostgreSQL, Docker, Streamlit, REST APIs, Git, GitHub, Simulink
Languages: Persian (Native), English (C1 - IELTS 7), Azerbaijani (C1), Italian (A2 - learning)

CERTIFICATIONS

Building with the Claude API - Anthropic | May 2026
SpaceCert - European Space Foundation | Sep 2025
ERC Space & Robotics Industry Standard - European Rover Challenge | Sep 2024